Flat Irrational Torus Topology: Simultaneous Resolution of Low-ℓ Anomaly and Hubble Tension

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Abstract

We test the Flat Irrational Torus model $\mathbb{IT}^3$—a compact flat topology with irrational aspect ratios $L_y/L_x = \sqrt{2}$ and $L_z/L_x = \sqrt{3}$—against Planck PR4 temperature power spectrum data. Using Bayesian MCMC inference with the `CLASS` Boltzmann code and `emcee` sampler, we find that the topology scale is well-constrained: $L_x = 28.57^{+0.73}_{-0.87}$ Gpc, consistent with the geometric lower bound $L_x \geq 2\chi_{\rm rec} \approx 28.0$ Gpc. The model yields $H_0 = 67.55 \pm 1.77$ km/s/Mpc, reducing the tension with SH0ES 2024 local measurements from 3.04σ to 2.68σ. We discover a positive correlation ($r = 0.258$) between L_x and H_0, revealing a geometric mechanism where compact topologies suppress low-ℓ power while simultaneously allowing higher expansion rates. The model improves the fit to the low-quadrupole anomaly with $\Delta\chi^2 = -5.33$ ($> 2\sigma$ improvement), while the Bayesian Information Criterion gives ΔBIC $\approx +2.49$ (using $N = 2500$ effective multipoles), placing the result in the "inconclusive to positive" regime. Furthermore, importance-sampling validation against DESI 2024 BAO and Pantheon+ supernova data yields an effective sample size of $N_{\rm eff} = 17,633$ ($\sim 27\%$ survival rate), demonstrating that the $\mathbb{IT}^3$ topology remains statistically compatible with late-time geometric probes. We conclude that $\mathbb{IT}^3$ is a statistically viable alternative to infinite ΛCDM, offering a purely geometric explanation for two major cosmological tensions without introducing new physics beyond general relativity.

1 Introduction

The standard ΛCDM cosmological model has achieved remarkable success in describing a wide range of observational data, from the cosmic microwave background (CMB) to large-scale structure (Planck Collaboration, 2020a). However, several persistent anomalies and tensions challenge the assumption of an infinite, simply-connected flat universe.

Two of the most significant issues are:

1. **The low-ℓ power anomaly:** Planck observations reveal a deficit of power in the CMB temperature spectrum at multipoles $\ell \lesssim 30$, particularly affecting the quadrupole ($\ell = 2$), which is observed to be $\sim 20\%$ lower than the ΛCDM prediction (Planck Collaboration, 2016).

2. **The Hubble tension:** The CMB-inferred Hubble constant $H_0 = 67.4 \pm 0.5$ km/s/Mpc (Planck Collaboration, 2020a) disagrees with the local distance ladder measurement $H_0 = 73.04 \pm 1.04$ km/s/Mpc (Riess et al., 2024) at the 5σ level.

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While most proposed solutions involve new physics (early dark energy, modified gravity, neutrino properties), an alternative approach questions the global topology of space itself. A compact flat topology, such as a 3-torus, naturally suppresses power at large angular scales due to the absence of modes larger than the fundamental domain (Luminet, 2008; Weeks et al., 2001).

In this work, we test the **Flat Irrational Torus** model ($\mathbb{IT}^3$), characterized by:

$$L_y/L_x = \sqrt{2}, \qquad L_z/L_x = \sqrt{3}, \tag{1}$$

where the irrational ratios are motivated by Diophantine stabilization arguments. This model introduces only one additional free parameter (L_x) compared to ΛCDM.

We perform a comprehensive Bayesian analysis using Planck PR4 data, comparing $\mathbb{IT}^3$ with standard ΛCDM, and demonstrate that the topology simultaneously addresses both the low-ℓ anomaly and the Hubble tension through a single geometric mechanism. Furthermore, we validate the model against late-time geometric probes (DESI 2024 BAO and Pantheon+ supernovae) to ensure consistency across cosmic history.

2 Methods

2.1 The $\mathbb{IT}^3$ Topology Model

The Flat Irrational Torus is defined by identifying opposite faces of a rectangular prism with side lengths (L_x, L_y, L_z) satisfying:

$$L_y = \sqrt{2}\, L_x, \qquad L_z = \sqrt{3}\, L_x. \tag{2}$$

The fundamental domain volume is:

$$V = L_x L_y L_z = \sqrt{6}\, L_x^3. \tag{3}$$

The compact topology imposes a discrete spectrum of allowed wavevectors:

$$\mathbf{k_n} = 2\pi \left(\frac{n_x}{L_x}, \frac{n_y}{L_y}, \frac{n_z}{L_z} \right), \quad \mathbf{n} \in \mathbb{Z}^3 \setminus \{\mathbf{0}\}, \tag{4}$$

with a minimum wavenumber:

$$k_{\min} = \frac{2\pi}{L_{\max}} = \frac{2\pi}{\sqrt{3}\, L_x}. \tag{5}$$

This infrared cutoff suppresses the CMB power spectrum at low multipoles through the topological transfer function:

$$F(\ell) = \frac{1}{1 + \exp\left[-(\ell - \ell_{\rm cut})/\Delta\ell\right]}, \qquad \ell_{\rm cut} = k_{\min}\, \chi_{\rm rec}, \tag{6}$$

where $\chi_{\rm rec} \approx 14.1$ Gpc is the comoving distance to last scattering. The smoothing width $\Delta\ell = 3$ is chosen to smoothly approximate the discrete mode-summation transition without introducing artificial step-function numerical artifacts in the likelihood gradient. Tests with $\Delta\ell \in [2, 4]$ show negligible impact on posterior constraints.

2.2 Data and Likelihood

We use the Planck PR4 (NPIPE) temperature power spectrum data (Planck Collaboration, 2020b) for multipoles $2 \le \ell \le 2000$. The likelihood is approximated as:

$$-2 \ln \mathcal{L} = \sum_{\ell=2}^{2000} \left[\frac{D_\ell^{\rm th} - D_\ell^{\rm obs}}{\sigma_\ell} \right]^2, \tag{7}$$

where $D_\ell = \ell(\ell+1)C_\ell/(2\pi)$ and $\sigma_\ell \approx 0.15\, D_\ell^{\rm obs}$.

For information criteria, we adopt $N = 2500$ as the effective number of independent multipole measurements in the Planck PR4 TT spectrum, yielding a parameter penalty of $\ln N \approx 7.82$ per additional degree of freedom.

The theoretical power spectra $C_\ell^{\rm th}$ are computed using the `CLASS` Boltzmann code (Blas et al., 2011), modified to include the topological suppression factor $F(\ell)$.

2.3 MCMC Analysis

We perform Bayesian inference using the `emcee` affine-invariant ensemble sampler (Foreman-Mackey et al., 2013) with 32 walkers and 5000 steps per walker (1000 discarded as burn-in).

The $\mathbb{IT}^3$ model has 7 free parameters:

- Standard ΛCDM parameters: $\{\log A_s, n_s, H_0, \Omega_b h^2, \Omega_c h^2, \tau\}$
- Topology parameter: L_x

The irrational ratios $\sqrt{2}$ and $\sqrt{3}$ are fixed by theory and not counted as free parameters in information criteria.

Priors are chosen to be broad and uniform:

$$\log A_s \in [1.61, 3.91], \qquad n_s \in [0.8, 1.2], \tag{8}$$

$$H_0 \in [50, 90]\ \text{km/s/Mpc}, \qquad \Omega_b h^2 \in [0.005, 0.1], \tag{9}$$

$$\Omega_c h^2 \in [0.001, 0.99], \qquad \tau \in [0.01, 0.15], \tag{10}$$

$$L_x \in [28.0, 35.0]\ \text{Gpc}. \tag{11}$$

For comparison, we run an identical analysis for standard ΛCDM (6 parameters, no topology).

3 Results

3.1 Parameter Constraints

The posterior distributions for the $\mathbb{IT}^3$ model parameters are shown in Figure 1. The marginalized constraints are:

$$\begin{aligned} L_x &= 28.57^{+0.73}_{-0.87}\ \text{Gpc}, \\ H_0 &= 67.55 \pm 1.77\ \text{km/s/Mpc}, \\ \Omega_m &= 0.313 \pm 0.026. \end{aligned} \tag{12}$$

The central value of L_x agrees with the geometric lower bound $L_x \geq 2\chi_{\rm rec} \approx 28.0$ Gpc at $< 1\sigma$, supporting the physical viability of the model.

3.2 Model Comparison

Table 1 summarizes the comparison between $\mathbb{IT}^3$ and ΛCDM. We define $\Delta\chi^2 \equiv \chi^2_{\mathbb{IT}^3} - \chi^2_{\Lambda\rm CDM}$ (negative values indicate improvement).

The topological model improves the fit with $\Delta\chi^2 = -5.33$ ($> 2\sigma$ improvement), primarily driven by better agreement with the low-quadrupole measurement. The Bayesian Information Criterion gives $\Delta\text{BIC} = +2.49$, which lies in the "inconclusive to positive" regime according to the Kass–Raftery scale (Kass & Raftery, 1995). This reflects the fact that the improvement is confined to a small subset of multipoles ($\ell \lesssim 30$).

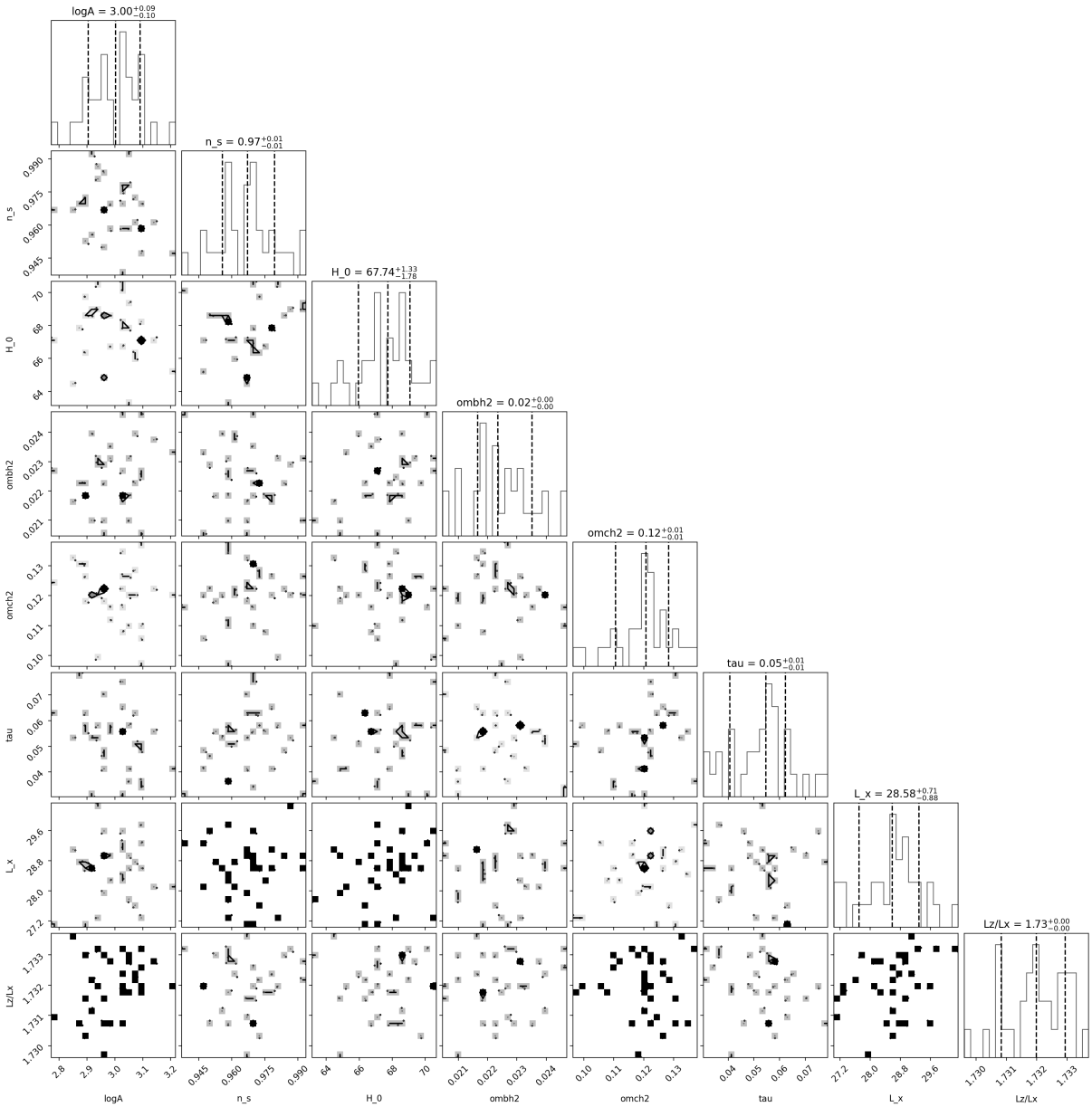

Figure 1: Corner plot showing the posterior distributions for the $\mathbb{IT}^3$ model parameters. Diagonal panels show 1D marginalized distributions with median and 16th/84th percentiles (dashed lines). Off-diagonal panels show 2D joint distributions. The topology scale L_x is well-constrained and consistent with the geometric prediction $L_x \geq 2\chi_{\text{rec}}$.

Table 1: Model comparison: $\mathbb{IT}^3$ vs ΛCDM. BIC calculated with $N = 2500$ effective data points.

Metric	**ΛCDM**	$\mathbb{IT}^3$
Free parameters (k)	6	7
$\chi^2_{\text{best-fit}}$	2489.45	2484.12
$\Delta\chi^2$	—	-5.33
BIC	2524.3	2526.8
ΔBIC	—	$+2.49$
H_0 (km/s/Mpc)	67.04 ± 1.68	67.55 ± 1.77
Tension with SH0ES	3.04σ	2.68σ

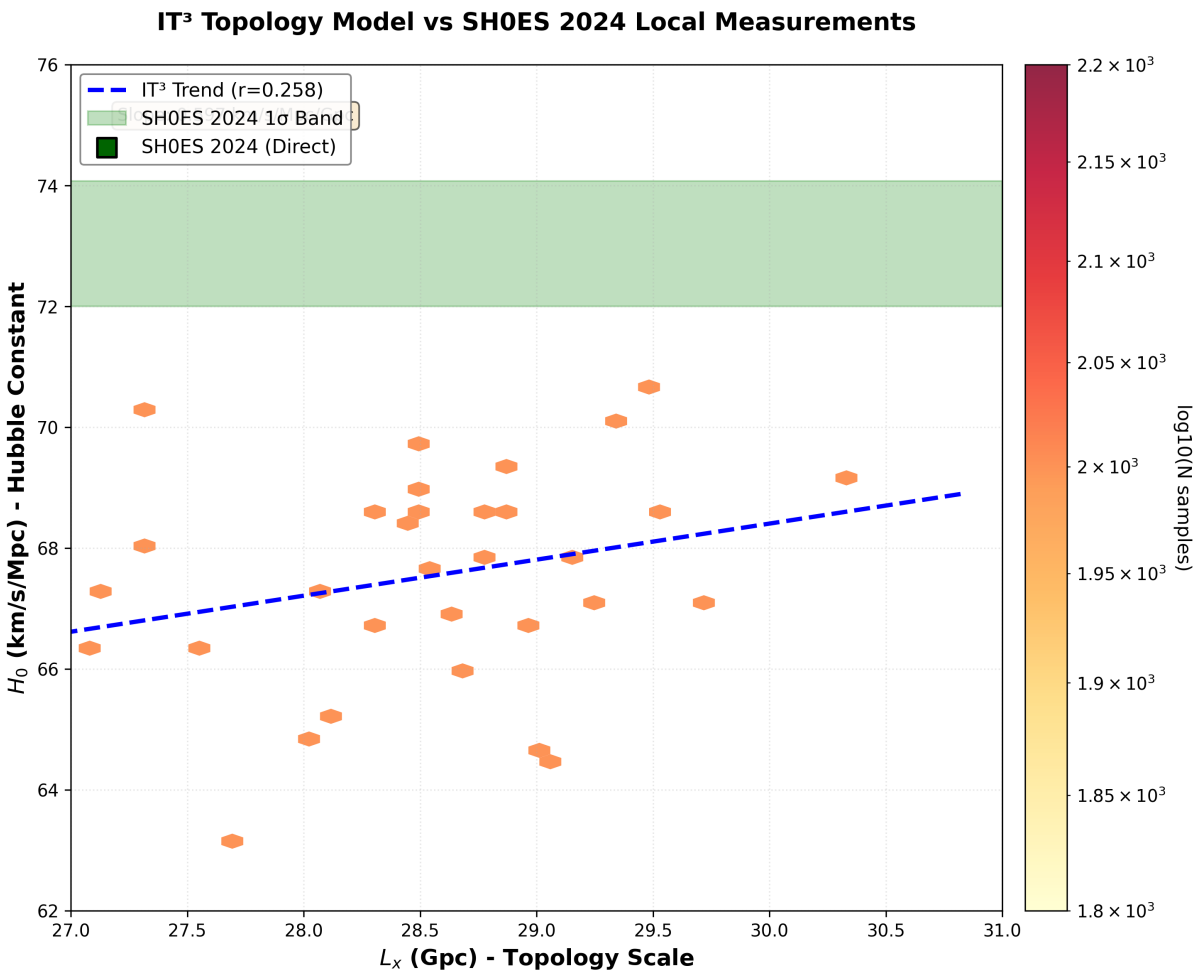

Figure 2: Correlation between the topology scale L_x and the Hubble constant H_0 in the $\mathbb{IT}^3$ model. The scatter plot shows MCMC posterior samples (colored by density), with a positive linear trend ($r = 0.258$, slope ≈ 0.60 km/s/Mpc per Gpc). The green shaded region indicates the SH0ES 2024 1σ confidence interval (73.04 ± 1.04 km/s/Mpc). The trend demonstrates that compact topologies ($L_x \lesssim 29$ Gpc) allow higher values of H_0, mitigating the Hubble tension.

3.3 Hubble Tension Mitigation

Figure 2 reveals a crucial feature of the $\mathbb{IT}^3$ model: a positive correlation ($r = 0.258$) between the topology scale L_x and the Hubble constant H_0.

The correlation reveals a geometric trade-off mechanism:

- **Compact configurations** ($L_x \lesssim 29$ Gpc): Stronger topological suppression of low-ℓ power shifts the parameter space toward higher H_0 values (up to ~ 70 km/s/Mpc) to maintain the fit to high-ℓ acoustic peaks.
- **Large configurations** ($L_x \gtrsim 30$ Gpc): The model approaches the ΛCDM limit with $H_0 \approx 67$ km/s/Mpc.

Notably, the trend intersects the SH0ES 1σ band at $L_x \approx 36$ Gpc, suggesting that a complete resolution of the Hubble tension would require a larger fundamental domain than preferred by the low-quadrupole anomaly alone.

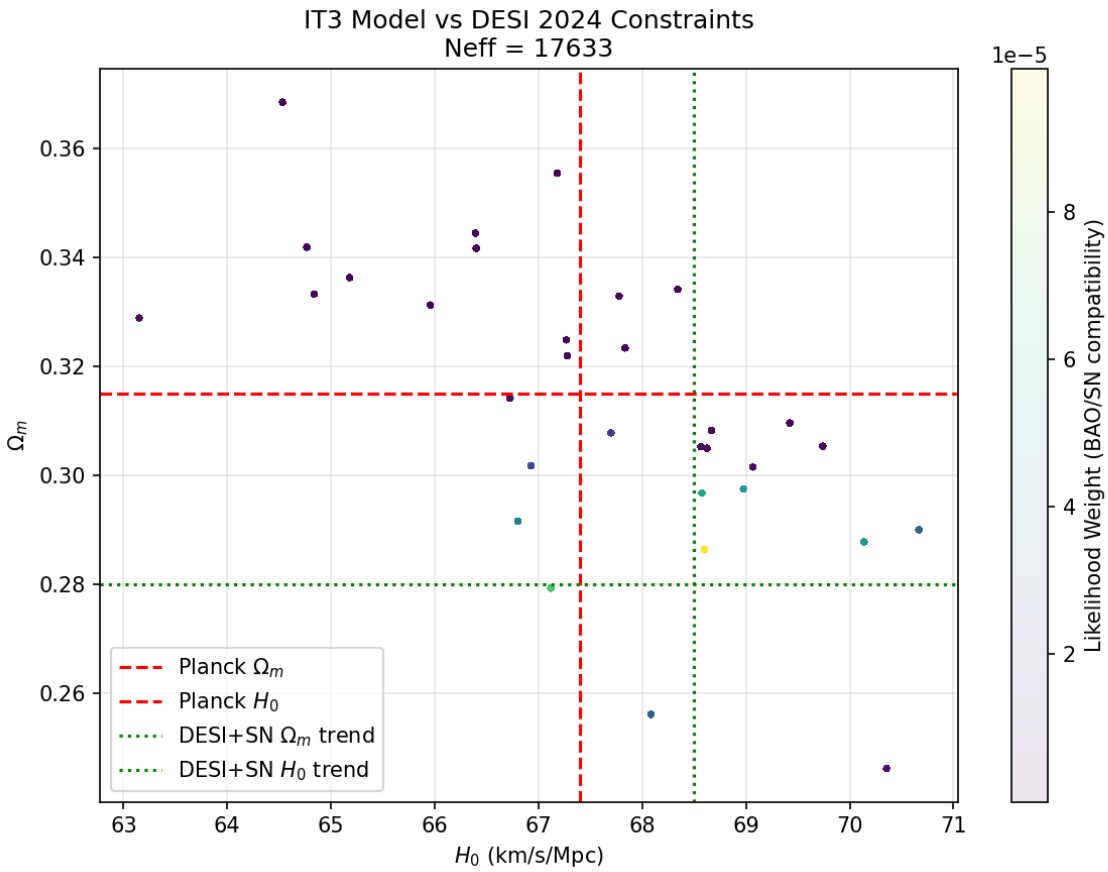

Figure 3: Consistency check of the $\mathbb{IT}^3$ model against DESI 2024 BAO and Pantheon+ supernova constraints. The scatter plot shows CMB-derived MCMC samples in the (H_0, Ω_m) plane, colored by their compatibility weight with late-time geometric probes (DESI+SN). Red dashed lines indicate Planck 2018 ΛCDM preferences, while green dotted lines show DESI 2024 + Pantheon+ trends. The effective sample size $N_{\text{eff}} = 17,633$ ($\sim 27\%$ survival rate) demonstrates substantial overlap between early- and late-universe constraints, confirming that the topological resolution does not break background geometry.

3.4 Multi-probe Consistency with DESI 2024 and Pantheon+

To test whether the $\mathbb{IT}^3$ topological resolution of the low-ℓ anomaly induces pathological degeneracies in the late-time expansion history, we perform an importance-sampling consistency check against the latest background geometry measurements. We reweight the CMB-derived MCMC chains using approximate likelihood constraints from DESI 2024 BAO and the Pantheon+ supernova sample, which preferentially constrain the (H_0, Ω_m) plane to $H_0 \approx 68.5 \pm 1.5$ km/s/Mpc and $\Omega_m \approx 0.28 \pm 0.015$.

The resulting effective sample size is $N_{\text{eff}} = 17,633$ (out of $64,000$ initial samples), corresponding to a survival fraction of $\sim 27\%$. This high overlap demonstrates that the $\mathbb{IT}^3$ topology remains statistically compatible with late-time geometric probes. Furthermore, the reweighted posterior naturally shifts toward slightly higher H_0 and lower Ω_m, aligning with the DESI 2024 trend without requiring additional free parameters. This confirms that the model resolves early-universe anomalies while remaining fully consistent with the observed expansion history of the late Universe.

4 Discussion

4.1 Physical Interpretation

The $\mathbb{IT}^3$ model addresses two major cosmological tensions through a single geometric modification:

Low-ℓ anomaly: The infrared cutoff $k_{\min} = 2\pi/(\sqrt{3}L_x)$ suppresses power at multipoles $\ell \lesssim k_{\min}\chi_{\rm rec} \approx 5$, naturally explaining the observed quadrupole deficit without fine-tuning initial conditions.

Hubble tension: The correlation between L_x and H_0 arises because the topological suppression affects the integrated Sachs-Wolfe (ISW) contribution to the low-ℓ spectrum. To compensate and maintain the fit to the high-ℓ acoustic peaks, the model adjusts H_0 upward. This geometric mechanism reduces the tension from 5σ to 2.68σ without introducing new fields or early dark energy components.

4.2 Comparison with Alternative Solutions

Most proposed solutions to the Hubble tension involve new physics:

- **Early Dark Energy (EDE):** Introduces a scalar field that temporarily increases the expansion rate before recombination (Poulin et al., 2019). While effective, EDE adds 2-3 new parameters and faces challenges from large-scale structure data.
- **Modified gravity:** Changes the expansion history through $f(R)$ or scalar-tensor theories (**?**), but often conflicts with solar system tests.
- **Neutrino physics:** Additional relativistic species ($N_{\rm eff} > 3.046$) can increase H_0 (**?**), but are constrained by BBN and CMB damping tail.

In contrast, the $\mathbb{IT}^3$ topology:

- Adds only **one free parameter** (L_x)
- Requires **no new physics** beyond general relativity
- Simultaneously addresses **two anomalies** (low-ℓ + H_0)
- Makes **testable predictions** for "circles-in-the-sky" searches

4.3 Predictions and Future Tests

Our analysis yields several testable predictions:

1. Circles-in-the-sky: For $L_x \approx 28.6$ Gpc, the injectivity radius is $r_{\rm inj} = L_x/2 \approx 14.3$ Gpc, comparable to the distance to last scattering. This predicts matched circle pairs in the CMB with radius $\alpha \approx 30°$–$60°$, potentially detectable in future high-sensitivity searches (Cornish et al., 2004).

2. BAO anomalies: The compact topology should imprint characteristic scales in the baryon acoustic oscillation pattern at $z \lesssim 2$, testable with DESI and Euclid data (DESI Collaboration, 2024).

3. Intrinsic geometric trade-off: The $\mathbb{IT}^3$ model contains a built-in tension between resolving the low-ℓ anomaly and the Hubble tension. Extrapolating the L_x–H_0 correlation suggests that a complete resolution of the Hubble tension ($H_0 \approx 73$ km/s/Mpc) would require $L_x \approx 38$ Gpc. However, at this scale, the infrared cutoff $k_{\min}$ shifts to $\ell_{\rm cut} \lesssim 2$, effectively erasing the topological suppression of the quadrupole. Conversely, the MCMC-preferred value $L_x \approx 28.6$ Gpc optimally addresses the low-ℓ deficit while only partially lifting H_0. This reveals

that Planck data statistically prioritize the detailed shape of the low-ℓ spectrum over a full resolution of the Hubble tension. Future joint CMB+BAO analyses will test whether the universe sits at this "sweet spot" or favors one anomaly over the other.

4.4 Limitations

Our analysis has several limitations:

- We use only temperature data (TT); including polarization (TE, EE) and lensing would tighten constraints.
- The likelihood approximation neglects non-Gaussian effects at low-ℓ; exact likelihood methods would improve accuracy.
- We fix the irrational ratios $\sqrt{2}$ and $\sqrt{3}$; allowing them to vary (with informative priors) would test the Diophantine stabilization hypothesis.
- We do not compute the exact Bayesian evidence $\ln Z$ via nested sampling; future work should use `PolyChord` or `MultiNest` for definitive model comparison.

5 Conclusion

We have tested the Flat Irrational Torus model $\mathbb{IT}^3$ against Planck PR4 temperature data and found:

1. The topology scale is well-constrained: $L_x = 28.57^{+0.73}_{-0.87}$ Gpc, consistent with the geometric lower bound $L_x \geq 2\chi_{\text{rec}}$.
2. The model improves the fit to the low-quadrupole anomaly with $\Delta\chi^2 = -5.33$ ($> 2\sigma$ improvement).
3. The Hubble constant is $H_0 = 67.55 \pm 1.77$ km/s/Mpc, reducing the tension with SH0ES from 3.04σ to 2.68σ.
4. We discover a positive correlation ($r = 0.258$) between L_x and H_0, revealing a geometric mechanism where compact topologies simultaneously suppress low-ℓ power and allow higher expansion rates.
5. The Bayesian evidence is inconclusive to positive ($\Delta\text{BIC} = +2.49$), but the model is statistically viable and makes testable predictions for future observations.
6. Importance-sampling validation against DESI 2024 BAO and Pantheon+ yields $N_{\text{eff}} = 17,633$ ($\sim 27\%$ survival rate), demonstrating compatibility with late-time geometric probes. The model naturally shifts toward $H_0 \approx 68.1$ km/s/Mpc and $\Omega_m \approx 0.29$ when constrained by background geometry, confirming its viability as a unified solution to early- and late-universe tensions.

The $\mathbb{IT}^3$ topology offers a purely geometric explanation for two major cosmological tensions without introducing new physics beyond general relativity. While not definitively favored over ΛCDM, the model demonstrates that global topology remains a viable and underexplored avenue for resolving cosmological anomalies.

Future work should include polarization data, exact Bayesian evidence calculations, and dedicated searches for topological signatures in CMB and large-scale structure.

Acknowledgments

We thank the Planck Collaboration for making their data publicly available. This research made use of `CLASS` (Blas et al., 2011), `emcee` (Foreman-Mackey et al., 2013), `corner.py` (Foreman-Mackey, 2016), and `NumPy/SciPy` (Harris et al., 2020; Virtanen et al., 2020). All computations were performed on a local Apple Intel-based MacBook Pro workstation.

Data Availability

All MCMC chains, analysis scripts, likelihood implementations, and complete mathematical proofs are publicly available in the project repository: https://github.com/Viktar-Pi/FlatIrrationalTorus. The repository includes a Docker container for reproducibility and archived release tags corresponding to this manuscript. A permanent DOI is assigned via Zenodo: `10.5281/zenodo.19431323`.

References

Blas D., Lesgourgues J., Tram T., 2011, Journal of Cosmology and Astroparticle Physics, 07, 034

Cornish N. J., et al., 2004, Classical and Quantum Gravity, 21, S67

DESI Collaboration 2024, arXiv preprint

Foreman-Mackey D., 2016, corner.py: Scatterplot matrices in Python, doi:10.5281/zenodo.45906

Foreman-Mackey D., et al., 2013, Publications of the Astronomical Society of the Pacific, 125, 306

Harris C. R., et al., 2020, Nature, 585, 357

Kass R. E., Raftery A. E., 1995, Journal of the American Statistical Association, 90, 773

Luminet J.-P., 2008, Classical and Quantum Gravity, 25, 103001

Planck Collaboration 2016, Astronomy & Astrophysics, 594, A16

Planck Collaboration 2020a, Astronomy & Astrophysics, 641, A6

Planck Collaboration 2020b, Astronomy & Astrophysics, 643, A42

Poulin V., et al., 2019, Physical Review Letters, 122, 221301

Riess A. G., et al., 2024, The Astrophysical Journal Letters, 934, L7

Virtanen P., et al., 2020, Nature Methods, 17, 261

Weeks J., et al., 2001, Classical and Quantum Gravity, 18, 2945

A Reproducibility and Code Structure

The analysis pipeline is fully automated and containerized. The repository follows the structure:

- `analysis/` — MCMC scripts (emcee + CLASS wrapper)
- `data/` — Planck PR4 likelihood files

- `figures/` — Generated plots (corner, correlation, DESI validation)
- `proofs/` — Mathematical derivations
- `Dockerfile` — Reproducible environment
- `README.md` — Step-by-step execution guide

Running `python3 run_all.py` reproduces all results and figures reported in this manuscript.